Synthesis Workshop

Virtual Poster

Efficient Syntheses of (S)- and (R)-β-Proline

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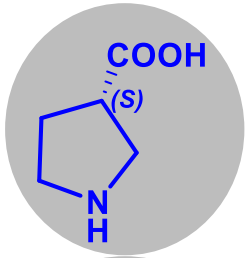
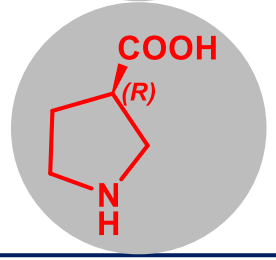
Introduction

β-Proline is a non-canonical β-amino acid

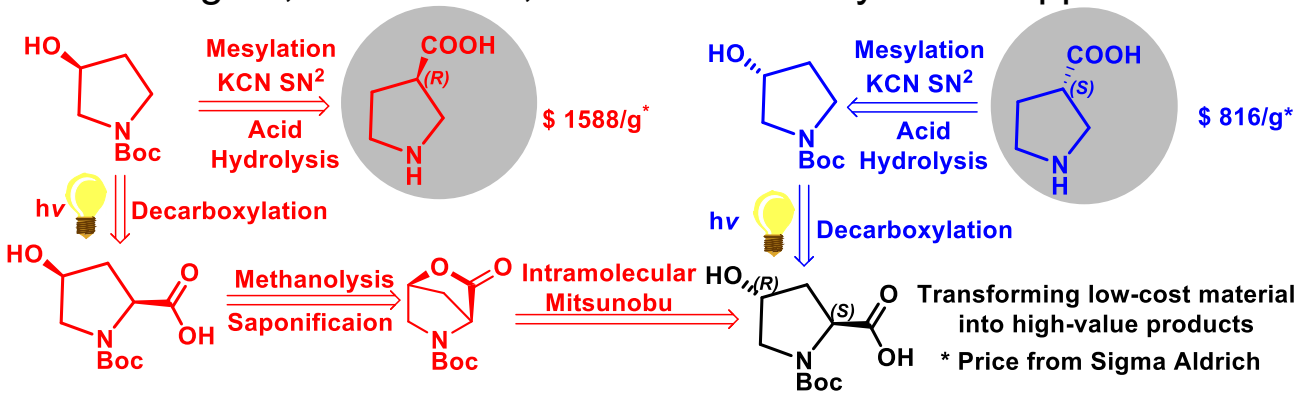




- Fundamental unit in construction of bioactive peptides and inhibitors
- Efficient organcatalyst in Aldol, Mannich and Michael addition reactions

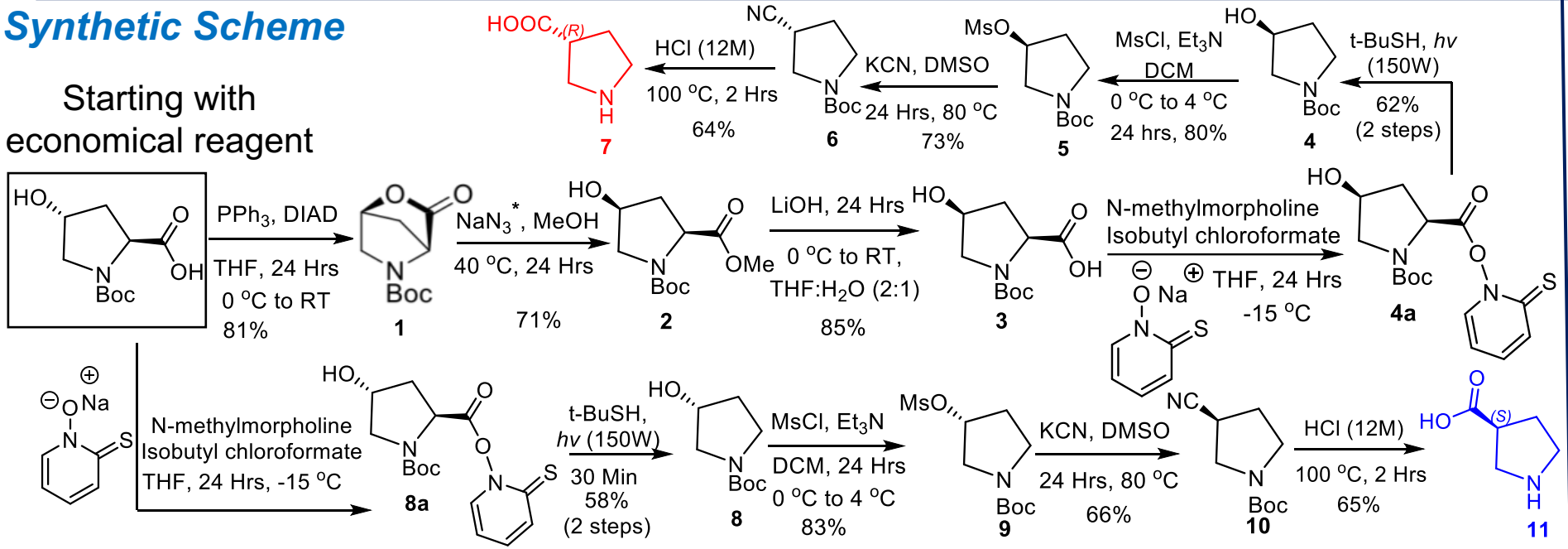
Approach

Enantiodivergent, economical, and metal-free synthetic approach:

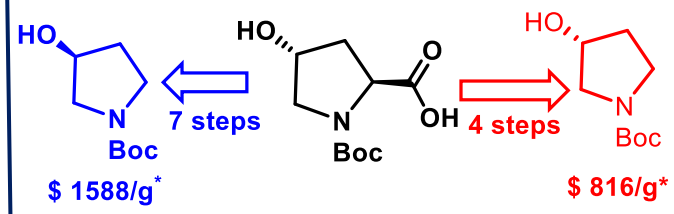


Synthetic Scheme

Starting with economical reagent



Conclusion



- Facile and short synthesis
- Economical, user-friendly

*Azide-assisted methanolysis to furnish methyl ester

S. D. Chandra et al. *Synthesis* **2023** 10.1055/a-2145-5470. *J. Am.Chem. Soc.* **2013**, *135*, 4333; *Tetrahedron* **1988**, *44*, 5479.